Scientific Skills Exercise

Making a Bar Graph and a Line Graph to Interpret Data

How Do Salinity and Competition Affect the Distribution of Plants in an Estuary? Field observations show that *Spartina patens* (salt marsh hay) is a dominant plant in salt marshes and *Typha angustifolia* (cattail) is a dominant plant in freshwater marshes. In this exercise, you will graph and interpret data from an experiment that examined the influence of an abiotic factor, salinity, and a biotic factor, competition, on the growth of these two species.

How the Experiment Was Done Researchers planted *S. patens* and *T. angustifolia* in salt marshes and freshwater marshes with and without neighboring plants. After two growing seasons (1.5 years), they measured the biomass of each species in each treatment. The researchers also grew both species in a greenhouse at six salinity levels and measured the biomass at each level after eight weeks.

Data from the Field Experiment (averages of 16 replicate samples)

	Average Biomass (g/100 cm^2)			
	Spartina patens		*Typha angustifolia*	
	Salt Marshes	Freshwater Marshes	Salt Marshes	Freshwater Marshes
With neighbors	8	3	0	18
Without neighbors	10	20	0	33

Data from the Greenhouse Experiment

Salinity (parts per thousand)	0	20	40	60	80	100
% maximum biomass (*S. patens*)	77	40	29	17	9	0
% maximum biomass (*T. angustifolia*)	80	20	10	0	0	0

Data from C. M. Crain et al., Physical and biotic drivers of plant distribution across estuarine salinity gradients, *Ecology* 85:2539–2549 (2004).

▲ *Spartina patens*

▲ *Typha angustifolia*

INTERPRET THE DATA

1. Make a bar graph of the data from the field experiment. (For additional information about graphs, see the Scientific Skills Review in Appendix D.) What do these data indicate about the salinity tolerances of *S. patens* and *T. angustifolia*?
2. What do the data from the field experiment indicate about the effect of competition on the growth of these two species? Which species was limited more by competition?
3. Make a line graph of the data from the greenhouse experiment. Decide which values constitute the dependent and independent variables, and use these values to set up the axes of your graph.
4. **(a)** In the field, *S. patens* is typically absent from freshwater marshes. Based on the data, does this appear to be due to salinity or competition? Explain your answer. **(b)** *T. angustifolia* does not grow in salt marshes. Does this appear to be due to salinity or competition? Explain your answer.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Too much light can also limit the survival of organisms. In some ecosystems, such as deserts, high light levels can increase temperature stress if animals and plants are unable to avoid the light or to cool themselves through evaporation (see Figure 40.12). At high elevations, the sun's rays are more likely to damage DNA and proteins because the atmosphere is thinner, absorbing less ultraviolet (UV) radiation. Damage from UV radiation, combined with other abiotic stresses, prevents trees from surviving above a certain elevation, resulting in the appearance of a tree line on mountain slopes **(Figure 52.23)**.

Rocks and Soil

In terrestrial environments, the pH, mineral composition, and physical structure of rocks and soil limit the distribution of plants and thus of the animals that feed on them, contributing to the patchiness of terrestrial ecosystems. The pH of soil can limit the distribution of organisms directly, through extreme acidic or basic conditions, or indirectly, by affecting the solubility of toxins and nutrients. Soil phosphorus, for instance, is relatively insoluble in basic soils and precipitates into forms unavailable to plants.

In a river, the composition of rocks and soil that make up the substrate (riverbed) can affect water chemistry, which in turn influences the resident organisms. In freshwater and marine environments, the structure of the substrate determines the organisms that can attach to it or burrow into it.

▼ **Figure 52.23 Alpine tree line in Banff National Park, Canada.** Organisms living at high elevations are exposed not only to high levels of ultraviolet radiation but also to freezing temperatures, moisture deficits, and strong winds. Above the tree line, the combination of such factors restricts the growth and survival of trees.